Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/26/2007
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Project Title:	Adaptation in Artificial Gravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
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COI Name (Institution):	DiZio, Paul (Brandeis University)		
Grant/Contract No.:	NAG9-01483		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The objective of the proposed research is to provide a technical base for evaluating the feasibility of a rotating "artificial gravity" environment for long-duration space missions. The project makes use of our unique rotating room facility to create an artificial gravity environment and a robotic manipulandum to partially simulate artificial gravity on the arm. The central concern is how to prevent the sensorimotor side effects of rotation caused by Coriolis, cross-coupled accelerations that are generated whenever the body moves in a rotating environment. Our specific aims are to assess: 1) whether it is possible for individuals to acquire dual adaptation to rotating and non-rotating environments allowing them to move between stationary and rotating environments without exhibiting errors in arm movement control; 2) whether adaptation of limb movement control per se in a rotating environment will transfer to wielding free objects and to operating machines that present constrained mechanical loads; 3) whether adaptation to movement-dependent, robotically simulated “Coriolis” forces can be used to preadapt astronauts to real inertial Coriolis forces such as they would encounter when moving about in a space vehicle rotating to generate artificial gravity. Together, the proposed		

	experiments will identify the upper limits of rotational velocities for artificial gravity environments in which limb movement control, tool handling and machine use can be readily adapted. This information is critical for developing design specifications for habitable artificial gravity environments in space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work on rotating artificial gravity environments has enlightened us about the presence of Coriolis and other inertial accelerations on the arms and legs during self-generated, voluntary movements in a normal terrestrial environment. Initial measurements have shown that Coriolis forces on the arm during natural turning and reaching movements are much greater than in our rotating room experiments, yet the hand reaches its goal accurately and the Coriolis forces are not noticed. Our current work on simultaneous adaptation to multiple force fields will help predict transfer of positive training effects and of negative aftereffects in a variety of man-machine interfaces on earth, including teleoperation.
Task Progress:	Specific Aim 1 We have found that when subjects are initially exposed to a rotating room the Coriolis forces generated when they attempt a reaching movement causes large errors in the endpoint and path of the movement. The dominant models of movement control had predicited there would be no endpoint errors. Similar errors occur for other types of movements – leg movements, head movements, whole body movements. We have found that 10-20 reaching movements in a rotating environment is sufficient to restore accurate reaching movements to a visual target. Earlier work had suggested that many orders of magnitude more movements would be required. The magnitude of initial reaching errors increases monotonically from room speeds of 5 to 30 rpm, and there is rapid, complete adaptation at all speeds. Our work has shown that adaptation is extraordinarily rapid if subjects are given a spatial goal or target to attain rather than just make random arm movements. We have also found that if subjects in a rotating environment stand passively, they do not show post-rotation postural aftereffects. However, if they voluntarily sway they rapidly adapt to the Coriolis forces generated by their movements and show large aftereffects post-rotation. The key to postural adaptation is - as with the arm – having spatial goals. We have now found that subjects who have adapted to one speed of rotation can be switched to other rates of rotation and maintain good accuracy. Once they are exposed to a new rotation speed their first reach is in error and subsequent reaches are again accurate. It is as if the first reach serves as a probe keying the CNS to the rotation rate and allowing corrective adjustments to be made. Specific Aim 2 We now have strong evidence that when subjects make reaching movements during passive body rotation that the way in which they adapt depends on the presence or absence of contact cues on the hand. If subjects reach with their hand free, then normal movement paths and accuracy will be regained within 20 movements. However, if the adapted subjects then reach while holding and object, renewed errors will result. By contrast, if subjects initially adapt while reaching and holding an object, they do not overcompensate when reaching without it. This means the CNS identifies some of the Coriolis forces during the initial reaches as having arisen from the object and some from the arm itself. The component of adaptation is tuned out when the subject reaches without the object. This decomposition of contacting and non-contacting forces on the arm must underlie context specific adaptation of tool use and self-calibration. Specific Aim 3 We studied adaptation to Coriolis force perturbations of reaching movements in a rotating environment when subjects reached freely to targets or when they reached to them while grasping a robotic manipulandum that was deactivated. After adaptation was achieved we then looked for transfer to when they reached grasping the deactivated robotic manipulandum or when they reached with the free hand. Subjects who initially reached holding the manipulandum showed little transfer to the free hand. This means they attributed the perturbation during the initial movements to the manipulandum not to properties of the environment or of the hand. Subjects who initially reached with the free hand showed retention of adaptation when reaching while holding the manipulandum. Such asymmetric transfer emphasizes the critical role of contact cues on the hand in determining the characteristics of adaptation, whether it is hand or tool dependent. Further experiments are necessary to determine if contact cues can be simulated – and in some conditions masked – to allow predictable manipulations of the adaptive state. This is key if robotic manipulanda are to be useful for pre-adapting astronauts to the Coriolis forces associated with rotation.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
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